Lab 08

ENVX2001 Applied Statistical Methods

Contents

Welcome to week 8!

This week's lab will give you some practice improving models by selecting the most useful predictors.

This will prepare you for next week's content where we will use models for prediction.

Learning outcomes

This module links to the following unit learning outcomes:

- LO3. demonstrate proficiency in the use or the statistical programming language R to apply an ANOVA and fit regression models to experimental data

- LO5. interpret the output and understand conceptually how its derived of a regression, ANOVA and multivariate analysis that have been calculated by R
- LO6. write statistical and modelling results as part of a scientific report

In this lab, we will learn how to:

1. Fit multiple linear regression models and interpret associated indicators of collinearity 2. Draw upon various methods to determine whether a full or reduced model is more suitable

Specific goals

By the end of this lab, you should be able to:

- ☐ Fit a multiple linear regression model
- ☐ Interpret VIF plots and output to determine whether collinearity is a concern in your model
- ☐ Remove non-significant predictors to produce a more parsimonious model
- ☐ Compare a full and reduced model using an F-test
- ☐ Run and interpret stepwise regression and associated output
- ☐ Compare a full and reduced model using the AIC
- ☐ Use the results of model comparisons to select the best model for your data

Preparation

Please work on these exercises by creating your own Quarto file. You are welcome to write your responses up in the style of a report.

Packages

This lab uses `tidyverse`, `readxl`, `moments`, `psych`, `performance`, `car`, and `gridExtra`. Install any you are missing by running the following **in the console**:

CODE

```
install.packages(c("car", "tidyverse", "readxl", "moments", "psych", "performance", "car",
"gridExtra"))
```

If a package is already installed, R will simply reinstall it — no harm done.

1. Variable selection for California streamflow (~ 45 mins)

Figure 1: Little Lakes Valley in the Eastern Sierra Nevada Mountains outside of Bishop, California. Source: Adobe Stock.

Last week you were investigating the relationship between precipitation at three sites (Lake Sabrina, Pine Creek, and Rock Creek) and stream runoff volume at a site near Bishop, California. You found that Lake Sabrina was not a significant predictor of runoff volume, but Pine Creek and Rock Creek were.

This week we will follow steps to see if we can produce a more parsimonious model. We will be focussing on the process rather than report writing, but we still encourage you to write up your conclusions as if it were for a report.

As a recap, the california_streamflow dataset you were analysing contains 43 years of annual precipitation measurements (in mm) taken at (originally) 6 sites in the Owens Valley in California. You will focus on only 3 variables.

The data is in the **california_streamflow.xlsx** file.

Variables are as follows:

- `lake_sabrina`: Precipitation recorded at Lake Sabrina (mm)
- `pine_creek`: Precipitation recorded at Big Pine Creek (mm),
- `rock_creek`: Precipitation recorded at Rock Creek (mm)
- `runoff_volume`: Stream runoff volume measured at a site near Bishop, California (ML/year). This will be the response variable.

Note the variables have already been log-transformed to increase normality of the residuals in the regressions.

Data source: Hollett, K.J., Danskin, F.M., McCaffrey, W.F., and Walti, S.L., 1991. Geology and water resources of Owens Valley, California. U.S. Geological Survey Water-Supply Paper 2370-H.

Exercise 1 - exploring full and reduced models

(i) Read in the data and investigate the relationship between each of your variables. Describe the relationships, supported by values (correlation coefficient) and plots.

(ii) Fit a multiple linear regression model to the data using all variables. Set `runoff_volume` as the response variable and `lake_sabrina`, `pine_creek` and `rock_creek` as the predictor variables. This will be known as the 'full model'.

(iii) Check the assumptions of your model. Do you need to transform any variables? What does the Variance Inflation Factor (VIF) tell you about the collinearity between predictor variables?

HINT: The VIF will appear as one of the plots in the `check_model()` function, but it can also be helpful to get the values. You can use the `vif()` function from the `car` package to get the VIF values for each predictor variable.

(iv) Make a note of the model parameters and fit statistics for the full model.

(v) Remove `lake_sabrina` from the model and run the model again. This will be known as the 'reduced model'.

(vi) Is the reduced model an improvement on the full model? How can you tell? What statistics would you look at to determine this?

Exercise 2 - which model is a significant improvement?

We can compare the models using the adjusted r^2, but we will need to use hypothesis testing to decide whether the reduced model is a significant improvement or not. Let's explore this by using the F-test to compare the full and reduced models.

(i) State the null and alternative hypotheses for the F-test between the full and reduced models.

(ii) Perform the F-test using the `anova()` function to compare the full and reduced models. What is the p-value? Based on this, what might we conclude to be the better model?

Exercise 3 - stepwise regression

Instead of removing one predictor at a time, there is a much quicker way involving the `step()` function in R. This function performs stepwise regression, which is an automated process of adding or removing predictors based on their significance in the model and the Akaike Information Criterion (AIC).

(i) When using the AIC, how do we select the better model?

(ii) Have a go at running the `step()` function on the full model. Which model does it select as the best model?

(iii) One final check: what are the AIC values for the full and reduced models? Which model has the lowest AIC value?

HINT: you can use the `AIC()` function to get the AIC values for each model. Insert the model name between the brackets.

(iv) Based on our tests, which model do you think is the best model to use for predicting runoff volume at Bishop, California? Why have you chosen this model?

Tip

Before moving on to the next section, take a short break and stretch your legs.

2. Variable selection for plant aridity (~ 45 mins)

Figure 2: Image of invasive capeweed. Source: Adobe Stock.

This dataset examines the adaptation of invasive capeweed (*Arctotheca calendula*) populations to aridity gradients across southern Australia. A common garden experiment was conducted from 2018-2019 with populations sourced from varying aridity levels, subjecting them to wet and dry treatments to measure traits related to drought escape, avoidance, and tolerance.

The dataset has been reduced to contain only the variables you will need to fit in the model. Furthermore, only the samples that received the wet treatment will be used for this analysis (n = 89 observations).

The dataset is in the *plant_aridity.csv* file.

Variables are as follows:

- `growth_rate`: growth rate measured as differences in # of rosette leaves between zero and eight weeks after transplanting

- `days_to_flower`: days to first flowering since transplant (days)
- `specific_leaf_area`: specific leaf area as [leaf area (cm2) / dry mass (g)]
- `root_shoot_ratio`: root-to-shoot ratio [root mass/shoot mass]
- `chlorogenic_acid`: concentration of chlorogenic acid (mg/g frwt sample)
- `rel_fitness`: Relative fitness of each plant, seed count divided by the mean seed count

`rel_fitness` will be the response variable in your models, and the other variables will be the predictor variables.

Data source: Uesugi, Akane (2022). Data from: Multivariate selection mediated by aridity predicts divergence of drought resistant traits along natural aridity gradients of an invasive weed [Dataset]. Dryad. https://doi.org/10.5061/dryad.tht76hf17

Paper: Carvalho C, Davis R, Connallon T, Gleadow RM, Moore JL, Uesugi A. Multivariate selection mediated by aridity predicts divergence of drought-resistant traits along natural aridity gradients of an invasive weed. New Phytol. 2022 May;234(3):1088-1100. doi: 10.1111/nph.18018.

Exercise 1 - initial analyses

(i) Write out the statistical model for a multiple linear regression and define it in terms of the data (i.e. relative fitness vs all other variables).

(ii) State the hypotheses you will be testing for the relationship between each variable and relative fitness.

(iii) Read in the data and conduct some exploratory data analysis upon your variables. Describe the spread and central tendency of each, supported by values (summary statistics) and plots.

(iv) Investigate the relationship between each of your variables. Describe the relationship, supported by values (correlation coefficient) and plots.

Hint: we are not only looking at relationship between response and predictors here, but also the relationships between predictor variables (signs of collinearity).

(v) Fit a multiple linear regression model to the data, with `rel_fitness` as the response variable and `growth_rate`, `days_to_flower`, `specific_leaf_area`, `root_shoot_ratio`, and `chlorogenic_acid` as the predictor variables. Label this model as `full_plant_model`.

(vi) Check the assumptions of your model. Do you need to transform any variables?

Hint 1: you can use the the `check_model()` function from the `performance` package to check the assumptions of your model.

Hint 2: If you are unsure about the collinearity plot, you can use the `vif()` function from the `car` package to get the VIF values for each predictor variable.

Hint 3: If you find that the assumptions are not met you can use log-transformations or other transformations as appropriate based on the distribution of your variables. If you are log-

> transforming a variable which contains negative or zero values you will need to add a constant, i.e. `log(x+1)`
>
> *(vii)* Interpret the model parameters and fit statistics. Are there any significant predictors of relative fitness? does the model explain much variation in relative fitness?
>
> *(viii)* Remove any highly non-significant predictors ($P > 0.1$) from the model and run the model again. This will be known as the 'reduced model'.
>
> *(ix)* Run a partial f-test to determine whether the reduced model an improvement on the full model. Which model is better?
>
> *(x)* Run stepwise regression using the `step()` function fitted to the full model. Which model does it select as the best model?
>
> *(viii)* Write a concluding statement on your findings.

Conclusion

Closing thoughts

You have done a great job working to improve and produce more parsimonious models.

As you may have noticed in this lab, models are rarely perfect, which is okay; it is our job to find the best way to explain variation in our response variables by exploring alternatives.

Keep up the great work and see you in the week 9 lecture for the last week of modelling!

If you have any questions, please approach your tutors or you are welcome to post to the Ed Discussion board.

Attribution